1 **(a)** $\frac{3}{8}$ correctly placed on the first branch for Blue **[B1]**

$\frac{5}{8}$ and $\frac{3}{8}$ correctly placed on all second branches **[B1]**

(b) $\frac{5}{8} \times \frac{5}{8}$ **[M1]**

$\frac{25}{64}$ **[A1]**

[Total Marks: 4]

2 **(a)** $\frac{3}{8}$ correctly placed for top branch (Red given Red) **[B1]**

$\frac{4}{8}$ correctly placed for bottom branches (Red given Blue, and Blue given Blue) **[B1]**

(b) $\frac{4}{9} \times \frac{5}{8}$ or $\frac{5}{9} \times \frac{4}{8}$ **[M1]**

$\frac{4}{9} \times \frac{5}{8} + \frac{5}{9} \times \frac{4}{8}$ **[M1]**

$\frac{5}{9}$ or equivalent (e.g. $\frac{40}{72}$) **[A1]**

[Total Marks: 5]

3 **(a)** $\frac{6}{9}$ on the Green branch from Green **[B1]**

$\frac{2}{9}$ and $\frac{7}{9}$ on the branches from Red **[B1]**

(b) $\frac{7}{10} \times \frac{6}{9}$ or $\frac{3}{10} \times \frac{2}{9}$ **[M1]**

$\frac{7}{10} \times \frac{6}{9} + \frac{3}{10} \times \frac{2}{9}$ **[M1]**

$\frac{48}{90}$ or $\frac{8}{15}$ **[A1]**

[Total Marks: 5]

4 **(a)** Intersection region = 6 **[B1]**

French only = 11, Spanish only = 8, outside = 5 **[B1]**

(b) **(i)** $\frac{6}{30}$ or $\frac{1}{5}$ **[B1]**

(ii) $\frac{11}{30}$ **[B1]**

(iii) $\frac{25}{30}$ or $\frac{5}{6}$ **[B1]**

[Total Marks: 5]

5 **(a)** Number of students studying both B and C: $(18 + 15) - (30 - 4) = 7$ **[M1]**

Fully correct Venn diagram: Only $B = 11$, $B \cap C = 7$, Only $C = 8$, Neither $= 4$ **[A1]**

(b) **(i)** $\frac{8}{30}$ or $\frac{4}{15}$ **[B1]**

(ii) $\frac{26}{30}$ or $\frac{13}{15}$ **[B1]**

(iii) $\frac{7}{18}$ **[B1]**

[Total Marks: 5]

6 **(a)** $\frac{3}{8}$ correctly placed for 1st ball Blue **[B1]**

$\frac{3}{8}$, $\frac{5}{8}$, and $\frac{3}{8}$ correctly placed on all three remaining 2nd ball branches **[B1]**

(b) $\frac{25}{64}$ **[B1]**

(c) $\left(\frac{5}{8} \times \frac{3}{8}\right) + \left(\frac{3}{8} \times \frac{5}{8}\right)$ **[M1]**

$\frac{30}{64}$ or $\frac{15}{32}$ **[A1]**

[Total Marks: 5]

7 **(a)** First counter branches: Red $\frac{5}{8}$, Blue $\frac{3}{8}$ **[B1]**

Second counter branches: Red $\frac{5}{8}$, Blue $\frac{3}{8}$ on both sets of branches **[B1]**

(b) $\frac{5}{8} \times \frac{5}{8} = \frac{25}{64}$ **[B1]**

(c) $\left(\frac{5}{8} \times \frac{3}{8}\right) + \left(\frac{3}{8} \times \frac{5}{8}\right)$ **[M1]**

$= \frac{30}{64}$ (or $\frac{15}{32}$) **[A1]**

[Total Marks: 5]

8 **(a)** $\frac{6}{9}$ and $\frac{3}{9}$ correctly placed on the top branches **[B1]**

$\frac{7}{9}$ and $\frac{2}{9}$ correctly placed on the bottom branches **[B1]**

(b) $\frac{7}{10} \times \frac{6}{9}$ or $\frac{3}{10} \times \frac{2}{9}$ **[M1]**

$\left(\frac{7}{10} \times \frac{6}{9}\right) + \left(\frac{3}{10} \times \frac{2}{9}\right)$ **[M1]**

$\frac{48}{90}$ or $\frac{8}{15}$ **[A1]**

[Total Marks: 5]

9 **(a)** Number studying both $= (18 + 15) - (30 - 5) = 8$ **[M1]**

Fully correct Venn diagram: B only $= 10$, $B \cap C = 8$, C only $= 7$, Outside $= 5$ **[A1]**

(b) **(i)** $\frac{10}{30}$ or $\frac{1}{3}$ **[B1]**

(ii) $\frac{25}{30}$ or $\frac{5}{6}$ **[B1]**

(c) Identifies total number of Chemistry students as denominator: 15 **[M1]**

$\frac{8}{15}$ **[A1]**

[Total Marks: 6]

10 **(a)** 8 in the intersection **[B1]**

16 in F only, 7 in C only, and 9 outside the circles **[B1]**

(b) Identifies 9 as the number of students playing neither **[M1]**

$\frac{9}{40}$ **[A1]**

(c) $16 + 8 + 7$ or $40 - 9$ **[M1]**

31 **[A1]**

[Total Marks: 6]

11 **(a)** $\frac{4}{7}$ correctly placed on first Blue branch **[B1]**

$\frac{3}{7}$ and $\frac{4}{7}$ correctly placed on all second branches **[B1]**

(b) $\frac{3}{7} \times \frac{3}{7}$ **[M1]**

$\frac{9}{49}$ **[A1]**

(c) $\frac{3}{7} \times \frac{4}{7} + \frac{4}{7} \times \frac{3}{7}$ or $2 \times \left(\frac{3}{7} \times \frac{4}{7}\right)$ **[M1]**

$\frac{24}{49}$ **[A1]**

[Total Marks: 6]

12 **(a)** First stage branches correctly labeled: $\frac{7}{12}$ for R and $\frac{5}{12}$ for B **[B1]**

Second stage branches from R correctly labeled: $\frac{6}{11}$ for R and $\frac{5}{11}$ for B **[B1]**

Second stage branches from B correctly labeled: $\frac{7}{11}$ for R and $\frac{4}{11}$ for B **[B1]**

(b) Multiplies probabilities along the R, R path: $\frac{7}{12} \times \frac{6}{11}$ **[M1]**

$\frac{42}{132}$ or $\frac{7}{22}$ **[A1]**

(c) Identifies and sums both correct combinations: $\left(\frac{7}{12} \times \frac{5}{11}\right) + \left(\frac{5}{12} \times \frac{7}{11}\right)$ or $2 \times \left(\frac{35}{132}\right)$ **[M1]**

$\frac{70}{132}$ or $\frac{35}{66}$ **[A1]**

[Total Marks: 7]

13 (a) $\frac{3}{8}$ placed correctly on the lower first branch **[B1]**

$\frac{5}{8}, \frac{3}{8}$ and $\frac{5}{8}, \frac{3}{8}$ placed correctly on the second branches **[B1]**

(b) $\frac{5}{8} \times \frac{5}{8}$ **[M1]**

$\frac{25}{64}$ **[A1]**

(c) $\frac{5}{8} \times \frac{3}{8}$ or $\frac{15}{64}$ **[M1]**

$2 \times \left(\frac{5}{8} \times \frac{3}{8}\right)$ **[M1]**

$\frac{15}{32}$ **[A1]**

[Total Marks: 7]

14 (a) All remaining rows completed correctly: row 1: $5, 6, 7$; row 2: $5, 6, 7, 8$; row 3: $5, 6, 7, 8, 9$; row 4: $5, 6, 7, 8, 9, 10$ **[B2]**

Award [B1] if at least 12 of the 18 missing entries are correct

(b) (i) $\frac{4}{24} = \frac{1}{6}$ **[B1]**

(ii) Prime outcomes $(2, 3, 5, 7)$ occur $1 + 2 + 4 + 4 = 11$ times **[M1]**

$\frac{11}{24}$ **[A1]**

(iii) Outcomes greater than 6 and even (8 and 10) occur $3 + 1 = 4$ times, giving $\frac{4}{24}$ **[M1]**

$\frac{1}{6}$ **[A1]**

(c) $\frac{1}{6} \times \frac{1}{6}$ **[M1]**

$\frac{1}{36}$ **[A1]**

[Total Marks: 9]

15 **(a)** $\dfrac{x}{x+4} \times \dfrac{x-1}{x+3}$ **[M1]**

$= \dfrac{x(x-1)}{(x+4)(x+3)} = \dfrac{x(x-1)}{x^2+7x+12}$ **[A1]**

(b) **(i)** $\dfrac{x(x-1)}{x^2+7x+12} = \dfrac{1}{3}$ **[M1]**

$3(x^2-x) = x^2+7x+12$ or $3x^2-3x = x^2+7x+12$ **[M1]**

$2x^2-10x-12 = 0 \implies x^2-5x-6 = 0$ **[A1]**

(ii) $(x-6)(x+1) = 0$ **[M1]**

$x = 6$ or $x = -1$ **[A1]**

Rejects $x = -1$ to conclude there are 6 blue counters **[B1]**

[Total Marks: 8]

16 **(a)** $\dfrac{5}{x+5} \times \dfrac{4}{x+4} = \dfrac{2}{15}$ **[M1]**

$20 \times 15 = 2(x+5)(x+4)$ or $300 = 2(x^2+9x+20)$ **[M1]**

$150 = x^2+9x+20$

$x^2+9x-130 = 0$ (No errors seen) [A1]

(b) $(x+13)(x-10) = 0$ **[M1]**

$x = 10$ **[A1]**

(c) Total counters $= 15$ **[B1]**

$\left(\dfrac{10}{15} \times \dfrac{5}{14}\right) + \left(\dfrac{5}{15} \times \dfrac{10}{14}\right)$ **[M1]**

$\dfrac{100}{210}$ or $\dfrac{10}{21}$ **[A1]**

[Total Marks: 8]

17 **(a)** Sets up correct equation for probability of two red balls: $\dfrac{x}{x+4} \times \dfrac{x-1}{x+3} = \dfrac{1}{3}$ **[M1]**

Cross-multiplies correctly to clear fractions: $3x(x-1) = (x+4)(x+3)$ **[M1]**

Expands and rearranges correctly with no errors:
$3x^2-3x = x^2+7x+12 \implies 2x^2-10x-12 = 0 \implies x^2-5x-6 = 0$ **[A1]**

(b) Factorises or solves the quadratic equation: $(x-6)(x+1) = 0$ **[M1]**

6 (and explicitly or implicitly rejects $x = -1$) **[A1]**

(c) Uses $x = 6$ to determine probabilities for one red and one blue ball: $\dfrac{6}{10} \times \dfrac{4}{9}$ **[M1]**

Accounts for both (R, B) and (B, R) combinations: $2 \times \left(\dfrac{6}{10} \times \dfrac{4}{9}\right)$ or $\dfrac{24}{90} + \dfrac{24}{90}$ **[M1]**

$\dfrac{48}{90}$ or $\dfrac{8}{15}$ **[A1]**

[Total Marks: 8]

18 **(a)** $\frac{3}{n} \times \frac{2}{n-1}$ **[M1]**

$\frac{6}{n(n-1)} = \frac{1}{15}$ **[M1]**

$n^2 - n = 90$

$n^2 - n - 90 = 0$ with no errors seen [A1]

(b) $(n-10)(n+9) = 0$ leading to $n = 10$ **[M1]**

7 **[A1]**

(c) $\frac{3}{10} \times \frac{7}{9}$ or $\frac{7}{10} \times \frac{3}{9}$ **[M1]**

$2 \times \left(\frac{3}{10} \times \frac{7}{9}\right)$ **[M1]**

$\frac{7}{15}$ or $\frac{42}{90}$ **[A1]**

[Total Marks: 8]

19 **(a)** $\frac{3}{n} \times \frac{2}{n-1}$ **[M1]**

$\frac{6}{n(n-1)} = \frac{1}{15}$ **[M1]**

$90 = n^2 - n$

$n^2 - n - 90 = 0$ [A1]

(b) $(n-10)(n+9) = 0$ **[M1]**

10 **[A1]**

(c) $n = 10$, giving 3 Red and 7 Blue balls **[B1]**

$2 \times \left(\frac{3}{10} \times \frac{7}{9}\right)$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ **[A1]**

[Total Marks: 8]

20 **(a)** $\frac{3}{n} \times \frac{2}{n-1} = \frac{1}{15}$ **[M1]**

$\frac{6}{n(n-1)} = \frac{1}{15}$ **[M1]**

$n^2 - n = 90 \implies n^2 - n - 90 = 0$ **[A1]**

(b) $(n-10)(n+9) = 0$ **[M1]**

$n = 10$ **[A1]**

(c) Number of Blue counters = 7 **[B1]**

$\left(\frac{3}{10} \times \frac{7}{9}\right) + \left(\frac{7}{10} \times \frac{3}{9}\right)$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ **[A1]**

[Total Marks: 8]

21 **(a)** $\frac{x}{x+3} \times \frac{x-1}{x+2} = \frac{5}{14}$ **[M1]**

$14x(x-1) = 5(x+3)(x+2)$ **[M1]**

$14x^2 - 14x = 5(x^2 + 5x + 6)$ **[M1]**

$14x^2 - 14x = 5x^2 + 25x + 30$ leading to $3x^2 - 13x - 10 = 0$ with no errors seen **[A1]**

(b) $(3x+2)(x-5) = 0$ or correct substitution into the quadratic formula **[M1]**

5 **[A1]**

(c) $\frac{5}{8} \times \frac{3}{7} + \frac{3}{8} \times \frac{5}{7}$ or $2 \times \left(\frac{5}{8} \times \frac{3}{7}\right)$ **[M1]**

$\frac{30}{56}$ or $\frac{15}{28}$ **[A1]**

[Total Marks: 8]

22 **(a)** $\frac{n}{n+6} \times \frac{n-1}{n+5} = \frac{1}{2}$ **[M1]**

$2n(n-1) = (n+6)(n+5)$ giving $2n^2 - 2n = n^2 + 11n + 30$ **[M1]**

$n^2 - 13n - 30 = 0$ **[A1]**

(b) $(n-15)(n+2) = 0$ **[M1]**

15 (rejecting $n = -2$ since $n > 0$) **[A1]**

(c) **(i)** $\left(\frac{15}{21} \times \frac{6}{20}\right) + \left(\frac{6}{21} \times \frac{15}{20}\right) = \frac{180}{420}$ **[M1]**

$\frac{3}{7}$ **[A1]**

(ii) $1 - \left(\frac{6}{21} \times \frac{5}{20}\right) = 1 - \frac{1}{14}$ **[M1]**

$\frac{13}{14}$ **[A1]**

(d) $3 \times \frac{15}{21} \times \frac{14}{20} \times \frac{6}{19}$ **[M2]**

$\frac{9}{19}$ **[A1]**

[Total Marks: 12]

23 **(a)** $\frac{2}{5}$ correctly placed on all three Tails branches **[B1]**

$\frac{3}{5}$ correctly placed on the remaining Heads branch **[B1]**

(b) $\frac{3}{5} \times \frac{2}{5} + \frac{2}{5} \times \frac{3}{5}$ or $2 \times \left(\frac{3}{5} \times \frac{2}{5}\right)$ **[M1]**

$\frac{12}{25}$ or 0.48 **[A1]**

[Total Marks: 4]

24 (a) 9 in the intersection of H and G **[B1]**

14, 10, and 7 in the correct regions **[B1]**

(b) $\frac{14}{40}$ or $\frac{7}{20}$ **[B1]**

(c) $\frac{9}{23} \times \frac{8}{22}$ **[M1]**

$\frac{72}{506}$ or $\frac{36}{253}$ **[A1]**

[Total Marks: 5]

25 (a) 6 placed correctly in the intersection **[B1]**

12 in F only, 8 in T only, and 4 outside the circles **[B1]**

(b) $\frac{6}{30}$ or $\frac{1}{5}$ or 0.2 **[B1]**

(c) $\frac{12}{30}$ seen as the probability for the first student **[M1]**

$\frac{12}{30} \times \frac{11}{29}$ **[M1]**

$\frac{132}{870}$ or $\frac{22}{145}$ **[A1]**

[Total Marks: 6]

26 (a) $2x + x + (x + 4) + 8 = 40$ **[M1]**

$4x + 12 = 40 \implies x = 7$ **[A1]**

(b) $\frac{32}{40}$ or $\frac{4}{5}$ **[B1]**

(c) Number of students playing Football only = 14 **[B1]**

$\frac{14}{40} \times \frac{13}{39}$ **[M1]**

$\frac{182}{1560}$ or $\frac{7}{60}$ **[A1]**

[Total Marks: 6]

27 (a) 10 in the intersection **[B1]**

14 in F only, 9 in T only, and 7 outside the circles **[B1]**

(b) $\frac{10}{40}$ or $\frac{1}{4}$ **[B1]**

(c) $\frac{10}{24} \times \frac{14}{23}$ or $\frac{14}{24} \times \frac{10}{23}$ **[M1]**

$\left(\frac{10}{24} \times \frac{14}{23}\right) + \left(\frac{14}{24} \times \frac{10}{23}\right)$ **[M1]**

$\frac{35}{69}$ or $\frac{280}{552}$ **[A1]**

[Total Marks: 6]

28 (a) Places 6 outside both circles **[B1]**

Calculates intersection: $32 + 21 + 6 - 50 = 9$ and places 9 in $H \cap G$ **[B1]**

Places 23 in H only and 12 in G only **[B1]**

(b) (i) $\frac{9}{50}$ **[B1]**

(ii) $\frac{38}{50}$ or $\frac{19}{25}$ **[B1]**

(c) Recognises the reduced sample space and uses probability without replacement: $\frac{9}{32} \times \frac{8}{31}$ **[M1]**

$\frac{72}{992}$ or $\frac{9}{124}$ **[A1]**

[Total Marks: 7]

29 (a) Intersection $n(G \cap H) = 8$ correctly placed **[B1]**

Remaining regions: $n(G \cap H') = 16$, $n(H \cap G') = 10$, outside region $= 6$ **[B1]**

(b) (i) $\frac{10}{40}$ (or $\frac{1}{4}$ or 0.25) **[B1]**

(ii) Identifies regions $16 + 8 + 6 = 30$ or calculates $1 - \frac{10}{40}$ **[M1]**

$\frac{30}{40}$ (or $\frac{3}{4}$ or 0.75) **[A1]**

(c) $\frac{24}{40} \times \frac{23}{39}$ **[M1]**

$= \frac{552}{1560}$ (or $\frac{23}{65}$ or 0.354) **[A1]**

[Total Marks: 7]

30 (a) 7 correctly placed in the intersection of T and G **[B1]**

17, 11, and 5 correctly placed in the remaining regions **[B1]**

(b) Identifies 17 students play tennis but not golf **[B1]**

$\frac{17}{40} \times \frac{16}{39}$ **[M1]**

$\frac{34}{195}$ or 0.174 (0.1743...) **[A1]**

(c) Denominator 18 seen **[M1]**

$\frac{7}{18}$ or 0.389 (0.3888...) **[A1]**

[Total Marks: 7]

31 (a) First branch $B = \frac{3}{10}$ **[B1]**

Top branch after R: $B = \frac{3}{9}$ **[B1]**

Bottom branches after B: $R = \frac{7}{9}$ and $B = \frac{2}{9}$ **[B1]**

(b) $\frac{7}{10} \times \frac{6}{9}$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ or 0.467 **[A1]**

(c) $\frac{7}{10} \times \frac{3}{9} + \frac{3}{10} \times \frac{7}{9}$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ or 0.467 **[A1]**

[Total Marks: 7]

32 (a) First pick: $\mathrm{P}(R) = \frac{5}{9}$ and $\mathrm{P}(Y) = \frac{4}{9}$ **[B1]**

Second pick: After R, $\mathrm{P}(R) = \frac{4}{8}$, $\mathrm{P}(Y) = \frac{4}{8}$; After Y, $\mathrm{P}(R) = \frac{5}{8}$, $\mathrm{P}(Y) = \frac{3}{8}$ **[B1]**

(b) $\mathrm{P}(RR) + \mathrm{P}(YY) = \left(\frac{5}{9} \times \frac{4}{8}\right) + \left(\frac{4}{9} \times \frac{3}{8}\right)$ **[M1]**

$\frac{20}{72} + \frac{12}{72} = \frac{32}{72} = \frac{4}{9}$ **[A1]**

(c) Method 1: $1 - \mathrm{P}(RR)$ or Method 2: $\mathrm{P}(RY) + \mathrm{P}(YR) + \mathrm{P}(YY)$ **[M1]**

$1 - \left(\frac{5}{9} \times \frac{4}{8}\right) = 1 - \frac{20}{72}$ **[M1]**

$\frac{52}{72} = \frac{13}{18}$ **[A1]**

[Total Marks: 7]

33 (a) First branches correct: $\mathrm{P}(R_1) = \frac{5}{8}$ and $\mathrm{P}(B_1) = \frac{3}{8}$ **[B1]**

Second branches after R_1 correct: $\mathrm{P}(R_2) = \frac{4}{7}$ and $\mathrm{P}(B_2) = \frac{3}{7}$ **[B1]**

Second branches after B_1 correct: $\mathrm{P}(R_2) = \frac{5}{7}$ and $\mathrm{P}(B_2) = \frac{2}{7}$ **[B1]**

(b) $\mathrm{P}(RR) + \mathrm{P}(BB) = \left(\frac{5}{8} \times \frac{4}{7}\right) + \left(\frac{3}{8} \times \frac{2}{7}\right)$ **[M1]**

$\frac{26}{56}$ or $\frac{13}{28}$ (or 0.464 or 0.4642...) **[A1]**

(c) $1 - \mathrm{P}(BB) = 1 - \left(\frac{3}{8} \times \frac{2}{7}\right)$ or $\left(\frac{5}{8} \times \frac{4}{7}\right) + \left(\frac{5}{8} \times \frac{3}{7}\right) + \left(\frac{3}{8} \times \frac{5}{7}\right)$ **[M1]**

$\frac{50}{56}$ or $\frac{25}{28}$ (or 0.893 or 0.8928...) **[A1]**

[Total Marks: 7]

34 (a) 3 placed correctly outside the circles **[B1]**

x, $24 - x$, and $18 - x$ placed correctly in the regions **[B1]**

(b) $(24 - x) + x + (18 - x) + 3 = 40$ or equivalent **[M1]**

5 **[A1]**

(c) $\frac{19}{40} \times \frac{13}{39}$ **[M1]**

$2 \times \left(\frac{19}{40} \times \frac{13}{39}\right)$ **[M1]**

$\frac{19}{60}$ or $\frac{494}{1560}$ or 0.316... **[A1]**

[Total Marks: 7]

35 (a) $18 + 14 + 3 - x = 30$ or $35 - x = 30$ **[M1]**

5 **[A1]**

(b) 5 placed correctly in the intersection **[B1]**

13 in F only, 9 in S only, and 3 outside the circles **[B1]**

(c) Identifies number of students studying only French is 13 **[M1]**

$\frac{13}{30} \times \frac{12}{29}$ **[M1]**

$\frac{156}{870}$ or $\frac{26}{145}$ **[A1]**

[Total Marks: 7]

36 (a) $\frac{3}{9}$ on the Red

Bluebranch[*B*1]

$\frac{7}{9}$ on the Blue

Redbranch[*B*1]

$\frac{2}{9}$ on the Blue

$Bluebranch[B1]$

(b) $\frac{7}{10} \times \frac{6}{9}$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ **[A1]**

(c) $\left(\frac{7}{10} \times \frac{3}{9}\right) + \left(\frac{3}{10} \times \frac{7}{9}\right)$ **[M1]**

$\frac{42}{90}$ or $\frac{7}{15}$ **[A1]**

[Total Marks: 7]

37 (a) 7 in the intersection **[B1]**

11, 8 and 4 in the correct regions **[B1]**

(b) $\frac{11}{30}$ **[B1]**

(c) $\frac{18}{30} \times \frac{17}{29}$ **[M1]**

$\frac{306}{870}$ or $\frac{51}{145}$ **[A1]**

(d) $\frac{7}{18} \times \frac{11}{17} + \frac{11}{18} \times \frac{7}{17}$ or $2 \times \left(\frac{7}{18} \times \frac{11}{17}\right)$ **[M2]**

Allow [M1] for one correct product $\frac{7}{18} \times \frac{11}{17}$

$\frac{154}{306}$ or $\frac{77}{153}$ **[A1]**

[Total Marks: 8]

38 (a) First disc Blue $= \frac{5}{12}$ **[B1]**

Second disc Blue (after Red) $= \frac{5}{11}$ **[B1]**

Second disc Red (after Blue) $= \frac{7}{11}$ and Blue (after Blue) $= \frac{4}{11}$ **[B1]**

(b) (i) $\frac{7}{12} \times \frac{6}{11}$ **[M1]**

$\frac{7}{22}$ **[A1]**

(ii) $\left(\frac{7}{12} \times \frac{5}{11}\right) + \left(\frac{5}{12} \times \frac{7}{11}\right)$ **[M2]**

$\frac{35}{66}$ **[A1]**

(c) $\left(\frac{7}{12}\right)^2 + \left(\frac{5}{12}\right)^2$ **[M1]**

$\frac{49}{144} + \frac{25}{144} = \frac{74}{144}$ **[M1]**

$\frac{37}{72}$ **[A1]**

[Total Marks: 11]

39 (a) $\frac{5}{n} \times \frac{4}{n-1} = \frac{2}{21}$ [M2]

(Award M1 for $\frac{5}{n} \times \frac{4}{n}$ or $\frac{5}{n} + \frac{4}{n-1}$)

$2(n^2 - n) = 420$ or $n(n-1) = 210$ **[M1]**

$n^2 - n - 210 = 0$ reached with no errors seen **[A1]**

(b) $(n-15)(n+14) = 0$ or correct substitution into quadratic formula **[M1]**

$n = 15$ **[A1]**

(c) $\frac{5}{15} \times \frac{10}{14}$ or $\frac{10}{15} \times \frac{5}{14}$ seen **[M1]**

Multiplies their probability by 2 (i.e. $2 \times \frac{50}{210}$) **[M1]**

$\frac{10}{21}$ or 0.476 (0.4761...) **[A1]**

[Total Marks: 8]

40 (a) P(Blue from A) $= \frac{x}{x+3}$ and P(Red from A) $= \frac{3}{x+3}$ **[B1]**

If Blue transferred to B, Box B has 5 Blue and 2 Red (total 7), so P(Blue from B) $= \frac{5}{7}$ **[B1]**

If Red transferred to B, Box B has 4 Blue and 3 Red (total 7), so P(Blue from B) $= \frac{4}{7}$ **[B1]**

Total probability $= \left(\frac{x}{x+3} \times \frac{5}{7}\right) + \left(\frac{3}{x+3} \times \frac{4}{7}\right) = \frac{5x+12}{7(x+3)}$ **[A1]**

(b) (i) $\frac{5x+12}{7(x+3)} = \frac{3}{4}$ **[M1]**

$4(5x+12) = 21(x+3) \implies 20x + 48 = 21x + 63$ **[M1]**

$x = 15$ **[A1]**

(ii) $\frac{\text{P(Red from A and Blue from B)}}{\text{P(Blue from B)}} = \frac{\frac{3}{18} \times \frac{4}{7}}{\frac{3}{4}} = \frac{\frac{1}{14}}{\frac{3}{4}} = \frac{2}{21}$ **[B1]**

[Total Marks: 8]

41 (a) $\frac{4}{x} \times \frac{3}{x-1}$ **[M1]**

$\dfrac{12}{x(x-1)} = \dfrac{1}{11}$ or $12 \times 11 = x(x-1)$ **[M1]**

$132 = x^2 - x \implies x^2 - x - 132 = 0$ **[A1]**

(b) $(x-12)(x+11) = 0$ or correct substitution into the quadratic formula **[M1]**

$x = 12$ **[A1]**

Number of blue discs $= 12 - 4 = 8$ **[A1]**

(c) $\dfrac{4}{12} \times \dfrac{8}{11}$ or $\dfrac{32}{132}$ **[M1]**

$\left(\dfrac{4}{12} \times \dfrac{8}{11}\right) \times 2$ **[M1]**

$\dfrac{64}{132}$ or $\dfrac{16}{33}$ **[A1]**

[Total Marks: 9]

42 (a) $\dfrac{4}{n} \times \dfrac{3}{n-1} = \dfrac{1}{14}$ **[M1]**

$\dfrac{12}{n^2 - n} = \dfrac{1}{14}$ **[M1]**

$168 = n^2 - n$ leading to $n^2 - n - 168 = 0$ with no errors seen **[A1]**

(b) $(n-14)(n+12) = 0$ or correct use of quadratic formula **[M1]**

$n = 14$ (rejecting $n = -12$) **[A1]**

Number of dark chocolates $= 14 - 4 = 10$ **[A1]**

(c) $\dfrac{4}{14} \times \dfrac{10}{13}$ or $\dfrac{10}{14} \times \dfrac{4}{13}$ **[M1]**

$2 \times \left(\dfrac{4}{14} \times \dfrac{10}{13}\right)$ **[M1]**

$\dfrac{80}{182}$ or $\dfrac{40}{91}$ **[A1]**

[Total Marks: 9]

43 (a) (i) Sets up equation $\dfrac{6}{n} \times \dfrac{5}{n-1} = \dfrac{1}{3}$ **[M1]**

Simplifies to $\dfrac{30}{n(n-1)} = \dfrac{1}{3}$ or $30 \times 3 = n(n-1)$ **[M1]**

$n^2 - n = 90$

$\mathrm{n}^2 - n - 90 = 0$ with no steps omitted [A1]

(ii) Factorises $(n-10)(n+9) = 0$ or uses quadratic formula correctly **[M1]**

Obtains $n = 10$ and $n = -9$ **[A1]**

Rejects negative root and states total number of sweets $= 10$ **[B1]**

(b) Calculates $P(\text{all 3 lemon}) = \dfrac{6}{10} \times \dfrac{5}{9} \times \dfrac{4}{8}$ **[M1]**

Identifies $P(\text{at least 1 orange}) = 1 - P(\text{all 3 lemon})$ **[M1]**

$= 1 - \dfrac{120}{720} = 1 - \dfrac{1}{6} = \dfrac{5}{6}$ (or 0.833) **[A1]**

[Total Marks: 9]

44 **(a)** Sets up correct initial probability equation: $\dfrac{4}{n} \times \dfrac{3}{n-1} = \dfrac{2}{15}$ **[M1]**

Simplifies equation: $\dfrac{12}{n(n-1)} = \dfrac{2}{15} \implies 2n(n-1) = 180$ **[M1]**

Expands and rearranges correctly to establish $n^2 - n - 90 = 0$ with no missing algebraic steps **[A1]**

(b) Factorises $(n-10)(n+9) = 0$ or uses quadratic formula correctly **[M1]**

Rejects negative root $n = -9$ and states $n = 10$ **[A1]**

(c) Identifies total number of chocolates $n = 10$, dark $= 4$, milk $= 6$ **[B1]**

Calculates probability of one correct arrangement (e.g., DD M): $\dfrac{4}{10} \times \dfrac{3}{9} \times \dfrac{6}{8} = \dfrac{72}{720}$ or $\dfrac{1}{10}$ **[M1]**

Multiplies single selection probability by 3 (accounting for DD M, D MD, M DD) **[M1]**

$\dfrac{216}{720}$ or $\dfrac{3}{10}$ (or 0.3) **[A1]**

[Total Marks: 9]

45 **(a)** $\dfrac{x}{x+4} \times \dfrac{x-1}{x+3} = \dfrac{5}{18}$ **[M1]**

$18x(x-1) = 5(x+4)(x+3)$ **[M1]**

$18x^2 - 18x = 5(x^2 + 7x + 12)$ **[M1]**

$13x^2 - 53x - 60 = 0$ obtained with no errors **[A1]**

(b) $(13x+12)(x-5) = 0$ or correct use of the quadratic formula [M2] (allow [M1] for one error in formula)

$x = 5$ or $x = -\dfrac{12}{13}$ **[A1]**

Selects $x = 5$ as the only valid solution for the number of apples **[B1]**

(c) Total apples $= 9$, green $= 5$, red $= 4$. Evaluates one branch: $\dfrac{5}{9} \times \dfrac{4}{8}$ or $\dfrac{4}{9} \times \dfrac{5}{8}$ **[M1]**

$\left(\dfrac{5}{9} \times \dfrac{4}{8}\right) + \left(\dfrac{4}{9} \times \dfrac{5}{8}\right)$ or multiplies one branch by 2 **[M1]**

$\dfrac{40}{72}$ or $\dfrac{5}{9}$ **[A1]**

[Total Marks: 10]